1.56 Equations of Lines in 3D - Vector Product Equations

Worked Solutions

1 a)

$$\left(\mathbf{r} - \begin{pmatrix}2\\8\\1\end{pmatrix}\right) \times \begin{pmatrix}5\\-2\\3\end{pmatrix} = \mathbf{0}.$$

1 b)

$$\left(\mathbf{r} - \begin{pmatrix}-10\\3\\-5\end{pmatrix}\right) \times \begin{pmatrix}7\\0\\-1\end{pmatrix} = \mathbf{0}.$$

2)

$$\left(\begin{pmatrix}2\\0\\3\end{pmatrix} - \begin{pmatrix}6\\3\\-2\end{pmatrix}\right) \times \begin{pmatrix}4\\3\\-5\end{pmatrix} = \begin{pmatrix}-4\\-3\\5\end{pmatrix} \times \begin{pmatrix}4\\3\\-5\end{pmatrix} = \begin{vmatrix}\mathbf{i} & \mathbf{j} & \mathbf{k}\\-4 & -3 & 5\\4 & 3 & -5\end{vmatrix}.$$

$$= ((-3)\times(-5) - 5\times 3)\mathbf{i} - ((-4)\times(-5) - 5\times 4)\mathbf{j} + ((-4)\times 3 - (-3)\times 4)\mathbf{k}.$$

$$= 0\mathbf{i} + 0\mathbf{j} + 0\mathbf{k} = \mathbf{0}.$$

$\therefore (2, 0, 3)$ lies on the line.

3 a)

$$\begin{pmatrix}5\\4\\-2\end{pmatrix} - \begin{pmatrix}-3\\0\\9\end{pmatrix} = \begin{pmatrix}8\\4\\-11\end{pmatrix}.$$

$$\therefore \left(\mathbf{r} - \begin{pmatrix}-3\\0\\9\end{pmatrix}\right) \times \begin{pmatrix}8\\4\\-11\end{pmatrix} = \mathbf{0}.$$

3 b)

$$\begin{pmatrix}-3\\9\\25\end{pmatrix} - \begin{pmatrix}-2\\-1\\-6\end{pmatrix} = \begin{pmatrix}-1\\10\\31\end{pmatrix}.$$

$$\therefore \left(\mathbf{r} - \begin{pmatrix}-2\\-1\\-6\end{pmatrix}\right) \times \begin{pmatrix}-1\\10\\31\end{pmatrix} = \mathbf{0}.$$

4)

$$\begin{pmatrix}4\\2\\1\end{pmatrix} = k\begin{pmatrix}2\\1\\0\end{pmatrix}.$$

$$4 = k(2) \Rightarrow k = 2.$$

$$2 = k(1) \Rightarrow k = 2.$$

$$1 = k(0) \Rightarrow k = 0.$$

Inconsistent values of k, so the lines are not parallel (or coincident).

5 a)

$$l_1 : \mathbf{r} = \begin{pmatrix}3\\-8\\4\end{pmatrix} + \lambda\begin{pmatrix}2\\1\\4\end{pmatrix}.$$

$$l_2 : \mathbf{r} = \begin{pmatrix}6\\5\\-3\end{pmatrix} + \mu\begin{pmatrix}-4\\3\\1\end{pmatrix}.$$

$$2 = k(-4) \Rightarrow k = -\frac{1}{2}.$$

$$1 = k(3) \Rightarrow k = \frac{1}{3}.$$

$$4 = k(1) \Rightarrow k = \frac{1}{4}.$$

Inconsistent values of k, so the lines are not parallel (or coincident).

$$3 + 2\lambda = 6 - 4\mu \Rightarrow 2\lambda + 4\mu = 3 \quad (1).$$

$$-8 + \lambda = 5 + 3\mu \Rightarrow \lambda - 3\mu = 13 \quad (2).$$

$$4 + 4\lambda = -3 + \mu \Rightarrow 4\lambda - \mu = -7 \quad (3).$$

Solve (1) and (2): $\lambda = \frac{61}{10}$ and $\mu = -\frac{23}{10}$. Check in (3):

$$4\left(\frac{61}{10}\right) - \left(-\frac{23}{10}\right) = \frac{267}{10} \neq -7.$$

So the lines do not intersect. l_1 and l_2 are skew lines.

5 b)

$$l_1 : \mathbf{r} = \begin{pmatrix}5\\4\\2\end{pmatrix} + \lambda\begin{pmatrix}-1\\-5\\1\end{pmatrix}.$$

$$l_2 : \mathbf{r} = \begin{pmatrix}3\\3\\4\end{pmatrix} + \mu\begin{pmatrix}6\\5\\-6\end{pmatrix}.$$

$$-1 = k(6) \Rightarrow k = -\frac{1}{6}.$$

$$-5 = k(5) \Rightarrow k = -1.$$

$$1 = k(-6) \Rightarrow k = -\frac{1}{6}.$$

Inconsistent values of k, so the lines are not parallel (or coincident).

$$5 - \lambda = 3 + 6\mu \Rightarrow -\lambda - 6\mu = -2 \quad (1).$$

$$4 - 5\lambda = 3 + 5\mu \Rightarrow -5\lambda - 5\mu = -1 \quad (2).$$

$$2 + \lambda = 4 - 6\mu \Rightarrow \lambda + 6\mu = 2 \quad (3).$$

Solve (1) and (2): $\lambda = -\frac{4}{25}$ and $\mu = \frac{9}{25}$. Check in (3):

$$-\frac{4}{25} + 6\left(\frac{9}{25}\right) = 2 \quad \text{as required.}$$

So the lines intersect. Substitute $\lambda = -\frac{4}{25}$ into l_1: The lines intersect at $\left(\frac{129}{25}, \frac{24}{5}, \frac{46}{25}\right)$.

6)

$$\cos\theta = \frac{\begin{pmatrix} 2 \\ -9 \\ 0 \end{pmatrix} \cdot \begin{pmatrix} 7 \\ 6 \\ 1 \end{pmatrix}}{\left|\begin{pmatrix} 2 \\ -9 \\ 0 \end{pmatrix}\right| \left|\begin{pmatrix} 7 \\ 6 \\ 1 \end{pmatrix}\right|} = \frac{2 \times 7 + (-9) \times 6 + 0 \times 1}{\sqrt{2^2 + (-9)^2 + 0^2}\sqrt{7^2 + 6^2 + 1^2}} = \frac{-40}{\sqrt{85}\sqrt{86}} = -\frac{40}{\sqrt{7310}}.$$

$$\theta = \arccos\left(-\frac{40}{\sqrt{7310}}\right) = 117.894\ldots^\circ.$$

Acute angle is $180^\circ - 117.894\ldots^\circ = 62.1^\circ$ (1 DP).

7 a)

$$l_1 : \mathbf{r} = \begin{pmatrix} 5 \\ -8 \\ 6 \end{pmatrix} + \lambda \begin{pmatrix} 3 \\ -7 \\ 1 \end{pmatrix}.$$

$$l_2 : \mathbf{r} = \begin{pmatrix} -9 \\ 10 \\ 3 \end{pmatrix} + \mu \begin{pmatrix} -13 \\ 1 \\ -1 \end{pmatrix}.$$

$$5 + 3\lambda = -9 - 13\mu \Rightarrow 3\lambda + 13\mu = -14 \quad (1).$$

$$-8 - 7\lambda = 10 + \mu \Rightarrow -7\lambda - \mu = 18 \quad (2).$$

$$6 + \lambda = 3 - \mu \Rightarrow \lambda + \mu = -3 \quad (3).$$

Solve (1) and (2): $\lambda = -\frac{5}{2}$ and $\mu = -\frac{1}{2}$. Check in (3):

$$-\frac{5}{2} - \frac{1}{2} = -3 \quad \text{as required.}$$

So the lines intersect. Substitute $\lambda = -\frac{5}{2}$ into l_1: The lines intersect at $\left(-\frac{5}{2}, \frac{19}{2}, \frac{7}{2}\right)$.

7 b)

$$\cos\theta = \frac{\begin{pmatrix}3\\-7\\1\end{pmatrix}\cdot\begin{pmatrix}-13\\1\\-1\end{pmatrix}}{\left|\begin{pmatrix}3\\-7\\1\end{pmatrix}\right|\left|\begin{pmatrix}-13\\1\\-1\end{pmatrix}\right|} = \frac{3\times(-13)+(-7)\times 1+1\times(-1)}{\sqrt{3^2+(-7)^2+1^2}\sqrt{(-13)^2+1^2+(-1)^2}}.$$

$$= \frac{-47}{\sqrt{59}\sqrt{171}} = -\frac{47}{\sqrt{10089}}.$$

$$\theta = \arccos\left(-\frac{47}{\sqrt{10089}}\right) = 117.899\ldots^\circ.$$

Acute angle is $180^\circ - 117.899\ldots^\circ = 62.1^\circ$ (1 DP).

7 c)

$$\overrightarrow{OA} = \begin{pmatrix}1\\0\\15\end{pmatrix}.$$

$$\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA} = \begin{pmatrix}5+3\lambda\\-8-7\lambda\\6+\lambda\end{pmatrix} - \begin{pmatrix}1\\0\\15\end{pmatrix} = \begin{pmatrix}4+3\lambda\\-8-7\lambda\\-9+\lambda\end{pmatrix}.$$

$\overrightarrow{AB}$ is perpendicular to l_1 so:

$$\begin{pmatrix}3\\-7\\1\end{pmatrix}\cdot\begin{pmatrix}4+3\lambda\\-8-7\lambda\\-9+\lambda\end{pmatrix} = 0.$$

$$\Rightarrow 3\times(4+3\lambda) - 7\times(-8-7\lambda) + 1\times(-9+\lambda) = 0.$$

$$\Rightarrow 12+9\lambda+56+49\lambda-9+\lambda = 0.$$

$$\Rightarrow 59\lambda+59 = 0.$$

$$\Rightarrow \lambda = -1.$$

$$\overrightarrow{OB} = \begin{pmatrix}5+3(-1)\\-8-7(-1)\\6+(-1)\end{pmatrix} = \begin{pmatrix}2\\-1\\5\end{pmatrix}.$$

So B has coordinates $(2,-1,5)$.

8 a)

$$\mathbf{r} = \begin{pmatrix}3\\-1\\0\end{pmatrix} + \lambda\begin{pmatrix}2\\9\\1\end{pmatrix}.$$

8 b)

$$\mathbf{r} = \begin{pmatrix}4\\3\\-2\end{pmatrix} + \lambda\begin{pmatrix}0\\2\\-3\end{pmatrix}.$$

9 a)

$$\frac{x-3}{4} = \frac{y-2}{-2} = \frac{z+9}{1} \quad \text{or} \quad \frac{x-3}{4} = \frac{2-y}{2} = z+9.$$

9 b)

$$\frac{x-8}{-2} = \frac{y+1}{-3} = \frac{z-0}{1} \quad \text{or} \quad \frac{8-x}{2} = \frac{y+1}{-3} = z.$$

9 c)

$$\frac{x-0}{1} = \frac{z-9}{-2}, \quad y = 2 \quad \text{or} \quad x = \frac{9-z}{2}, \quad y = 2.$$

10 a)

Vector Form:

$$\mathbf{r} = \begin{pmatrix} 3 \\ -2 \\ 4 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 4 \\ 2 \end{pmatrix}.$$

Cartesian Form:

$$x = 3 + \lambda \Rightarrow \lambda = x - 3.$$
$$y = -2 + 4\lambda \Rightarrow \lambda = \frac{y+2}{4}.$$
$$z = 4 + 2\lambda \Rightarrow \lambda = \frac{z-4}{2}.$$
$$\therefore x - 3 = \frac{y+2}{4} = \frac{z-4}{2}.$$

10 b)

Vector Form:

$$\mathbf{r} = \begin{pmatrix} 3 \\ -7 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} 0 \\ 4 \\ -5 \end{pmatrix}.$$

Cartesian Form:

$$x = 3.$$
$$y = -7 + 4\lambda \Rightarrow \lambda = \frac{y+2}{4}.$$
$$z = 0 - 5\lambda \Rightarrow \lambda = \frac{z}{-5}.$$
$$\therefore x = 3, \quad \frac{y+2}{4} = \frac{z}{-5}.$$